Data Integrity & Drift Detection Diagnostic Report

Overview

This report evaluates data integrity and drift within the deployed RAG-based system using audit log data. The analysis compares reference (baseline) and current production distributions to detect feature drift, assess its impact on model performance, and recommend appropriate interventions.

A total of 378 observations were analyzed. Statistical methods including Population Stability Index (PSI) and Kolmogorov–Smirnov (KS) tests were used to quantify drift across key features.

1. Features That Drifted Most

The features evaluated for drift revealed several important shifts in the system's performance metrics. The Retrieval Score demonstrated the highest level of drift among all features, recording a PSI value of 0.150 and a KS Statistic of 0.127 with a p-value of 0.0949. This places it firmly in the moderate drift category (typically defined as PSI between 0.1 and 0.2), suggesting a meaningful change in the distribution of retrieval quality scores compared to the reference period. Meanwhile, Query Length showed only minimal drift with a PSI of 0.085, indicating that the distribution of query lengths has remained relatively stable and consistent over time.

In terms of operational metrics, the Refusal Rate experienced a slight increase, rising from 0.630 in the reference dataset to 0.714 in the current period. This uptick may warrant monitoring but does not appear dramatic. On a positive note, the Groundedness Rate improved from 0.831 to 0.878, reflecting better performance in generating responses that are well-supported by retrieved information. In conclusion, while the Retrieval Score stands out as the primary area of moderate drift, the other features generally exhibited minimal or expected variations, with the improvement in groundedness being a particularly encouraging development.

2. Impact on Model Performance

Despite the observed moderate drift in retrieval scores, the overall system performance remains stable. The KS test p-value of 0.0949 indicates that the drift is not statistically significant at the conventional 0.05 level, meaning there is insufficient evidence to conclude a strong distributional shift. Additionally, the Groundedness Rate showed a clear improvement, increasing from 0.831 in the reference period to 0.878 in the current period. This suggests enhanced factual alignment and better-supported responses generated by the system. The Refusal Rate, on the other hand, experienced a slight increase from 0.630 to 0.714, which may reflect stricter filtering mechanisms or more cautious model behavior when handling uncertain or borderline queries.

These findings carry several important implications for the long-term health of the system. While the moderate drift in retrieval scores could potentially impact answer relevance if the trend continues unchecked, the concurrent improvement in groundedness appears to offset any negative effects for now. As a result, no immediate degradation in overall system quality has been observed. Ongoing monitoring of the retrieval component is recommended to prevent future issues, but the current balance of metrics supports continued system stability.

3. Retraining or Intervention Recommendations

Based on the analysis, the following actions are recommended in the short term. Monitoring of drift metrics should continue without immediate retraining, allowing the team to gather more data and observe trends. To ensure proactive management, clear alert thresholds should be established, including a PSI exceeding 0.20 for any feature, a significant drop in the groundedness rate, or a rapid increase in the refusal rate. These thresholds will help flag issues early before they affect overall system performance.

In the medium term, efforts should focus on investigating the root causes of the retrieval score drift. This includes analyzing potential changes in input query patterns and any shifts in the production data distribution. Additional metadata logging, such as query categories or context types, may be implemented to provide deeper insights into the observed changes. For the long term, a scheduled retraining pipeline should be developed and triggered automatically based on sustained PSI values above 0.20, statistically significant KS test results, or any measurable degradation in key performance metrics. This structured approach will help maintain system stability and quality over time.

4. Conclusion

The system shows no major drift requiring immediate intervention. While retrieval score drift is present at a moderate level, it has not negatively impacted overall model performance. In fact, the improvement in groundedness suggests stable or even enhanced response quality. The recommended approach is to continue monitoring with clearly defined thresholds and only initiate retraining when the drift becomes statistically significant or begins to affect key performance metrics.